

Differential Equations

A lightning review of partial differential equations.

We've seen the maxwell equations, the Schrödinger equation, etc.

We usually use separation of variables to solve ordinary differential equations.

Lets take a prototypical equation.

$$y''(x) + P(x)y' + Q(x)y = R(x) \quad (1)$$

We are worried about nonlinearities. $R(x)$ is a forcing term. If P, Q have a power series expansion that works around the point we care about, x_0 .

It is the point where we can make the equation legal without setting one of them to zero.

In many equations, this isn't true - a singular point.

There are also regular singular points, where we have

$(x - x_0)P(x)$ has a convergent series and $(x - x_0)^2Q(x)$ also has a convergent series. If this is the case, we can look for a series expansion of the solution functions, which we will do.

If these don't converge, it is a lot harder.

For example:

Bessel's equation looks like

$$x^2y'' + xy' + (x^2 - n^2)y = 0 \quad (2)$$

We have to put it in the nice form, so we divide by x^2

$$y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right)y = 0 \quad (3)$$

This has a singular point at $x = 0$. If we multiply $\frac{1}{x}$ by $(x - x_0)$, then we have a nice thing. Similarly, we have a nice part for $\left(1 - \frac{n^2}{x^2}\right)(x - x_0)^2$.

We have this centered at 0 where it would explode.

We define DDSHO

$$y'' + 2\beta y' + \omega_0^2 y = \frac{F(x)}{m} \quad (4)$$

where $\beta = \frac{b}{m}$.

what about

$$y'' + \frac{\alpha}{x}y' + \frac{\beta}{x^2}y = 0 \quad (5)$$

This is really stupid simple to solve. It is called an equidimensional equation, since everything has the same dimension.

Lets guess $y = x^p$.

This would become

$$p(p-1)x^{p-2} + \frac{\alpha}{x}px^{p-1} + \frac{\beta}{x^2}x^p = 0 \quad (6)$$

$$x^{p-2} [p(p-1) + p\alpha + \beta] = 0 \quad (7)$$

This is a quadratic equation to solve for the power of an equidimensional equation, which has two linearly independent solutions.

Another example:

$$(1-x^2)y'' - 2xy' + l(l+1)y = 0 \quad (8)$$

This has singular points at $x = \pm 1$. This is Legendre's equation from quantum mechanics, where we are solving for the polar angular equation (north to south pole).

1 Sin and Cos

Lets work all the way through an equation:

$$y'' + y = 0 \quad (9)$$

We have two linearly independent equations. Lets choose (arbitrary numbers) that are linearly independent:

$$\alpha(0) = 1 \quad \alpha'(0) = 0 \quad (10)$$

$$\beta(0) = 0 \quad \beta'(0) = 1 \quad (11)$$

We have to have two pieces of information to carry forwards (imagine a ball frozen in space - you need to know its position and velocity to time evolve it). We could use different slopes but would just get a phase shift relative to the sin and cos solutions. And the amplitudes don't matter, as long as they are orthogonal - so zero and something. Just 1 is a nice number.

By consequence of the original equation,

$$y''' + y' = 0 \quad (12)$$

let $\eta = y'$
 $\eta'' + \eta = 0$.

Then $\alpha'(x)$ has to be a linear combination of the two independent solutions, $c\alpha(x) + d\beta(x)$

$$\alpha'(0) = 0 = d\beta(0) \quad (13)$$

$$\alpha'(x) = d\beta(x) \quad (14)$$

Similarly,

$$\beta'(0) = 1 = f + 0g \quad (15)$$

$$\beta'(x) = \alpha + g\beta(x) \quad (16)$$

$$\beta''(x) = \alpha'(x) + g\beta'(x) \quad (17)$$

$$-\beta(x) = d\beta(x) + g\beta'(x) \quad (18)$$

$$\therefore g = 0 \quad (19)$$

$$(20)$$

We now have

$$\beta'(x) = \alpha(x) \quad (21)$$

$$\alpha'(x) = d\beta(x) \quad (22)$$

$$\alpha''(x) = -\alpha(x) = d\beta'(x) \quad (23)$$

At $x = 0$, $-1 = d$.

How do we prove $\alpha^2 + \beta^2 = 1$?

We can take a derivative and show that it is 0, then it's a constant.

$$2\alpha\alpha' + 2\beta\beta' = c \quad (24)$$

$$\text{recall that } d = -1, \quad (25)$$

$$\alpha = -\beta(x) \quad (26)$$

$$\beta' = \alpha(x) \quad (27)$$

$$\text{so the derivative is} \quad (28)$$

$$2\alpha(-\beta) + 2\beta(\alpha) = 0 \quad (29)$$

Therefore, $\alpha^2 + \beta^2 = 1$ is true everywhere.

Lets move on to

2 Wronskian

$$y'' = A(x)y' + B(x)y \quad (30)$$

Suppose that $f(x), g(x)$ solve the differential equation.

The Wronskian is $w = fg' - f'g$.

We can take the derivative: $\frac{dw}{dx}$. It smells faintly like the bilinear concomitant thing...

$$\frac{dw}{dx} = f'g' + fg'' - f''g - f'g' \quad (31)$$

$$= fg'' - f''g \quad (32)$$

$$(33)$$

But, each of the $A(x)$ and $B(x)$ are solutions. We can plug those in and see what we get.

$$\frac{dw}{dx} = f[Ag' + Bg] - [Af' + Bf]g \quad (34)$$

$$= A(x)[fg' - f'g] \quad (35)$$

This is the Wronskian itself: $A(x)w(x)$

in other words,

$$\frac{1}{w} \frac{dw}{dx} = A(x) \quad (36)$$

This is the logarithmic derivative,

$$\frac{d \ln(w)}{dx} = A(x) \quad (37)$$

$$\ln(w) = \int A(x) dx + C \quad (38)$$

$$W(x) = Ce^{\int A(x) dx} \quad (39)$$

So what? What can we use this for?

This is called Abel's formula. In essence, it is a way of getting a second solution if you already know the first.

Lets take a DE to play with:

$$x^2 y'' + 2xy' - 2y = 0 \quad (x > 0) \quad (40)$$

The first thing for us to show is that

$y_1(x) = x$ is a solution, and second is to use Abel's formula to find y_2 .

The first is easy to verify by just plugging it in.

Next:

$$y'' = - \underbrace{\frac{2}{x}}_{A(x)} y' + \frac{2}{x^2} y \quad (41)$$

$$w(x) = c e^{\int \frac{-2}{x} dx} = c x e^{-2}$$

$w = f g' = f' g$. We have one solution. We have

$$x y_2' - y_2 = \frac{C}{x^2} \quad (42)$$

We can now solve this:

$$x^2 \frac{d}{dx} \left(\frac{y_2}{x} \right) = \frac{C}{x^2} \quad (43)$$

$$d \left(\frac{y_2^2}{x} \right) = \frac{C}{x^4} dx \quad (44)$$

$$\frac{y_2}{x} = \int \frac{C}{x^4} dx = -\frac{C}{3} \frac{1}{x^3} + C' \quad (45)$$

$$y_2 = -\frac{C}{3} \frac{1}{x^2} + x C' \xrightarrow{0} 0 \quad (46)$$